Parichit Sharma

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Research Gate, Google Scholar, GitHub, Profile at HBKU, Qatar

Research Summary

My research advances **Data-Centric AI, Applied Machine Learning, and Data Science** with a unified vision of transforming complex scientific challenges through intelligent data processing. I pioneer **data expressiveness principles** that identify influential data points, achieving up to **70% reduction in training time and 83-86%** energy savings in clustering algorithms without sacrificing accuracy—fundamentally reimagining how we train machine learning (ML) models. My expertise spans **pattern identification across varying data-availability regimes**, from enhancing cellular discernability in high-dimensional **single cell** data to maximizing insights from limited **protein datasets**. My ML frameworks accelerate scientific discovery across **biology, materials science, and energy applications**, transforming complex datasets into actionable insights for next-generation materials discovery and cellular heterogeneity analysis.

Education

Ph.D. in Computer Science, **Minor in** Data Science, **GPA: 3.90/4.0**
Indiana University, Bloomington, IN, USA **2019 - 2025**

Advisor: Prof. Mehmet M. Dalkilic
Committee Members: Prof. Haixu Tang, Assoc. Prof. Prateek Sharma, Asst. Prof. Zoran Tiganj
Asst. Prof. Hasan Kurban

Master of Science in Computer Science, **GPA: 3.89/4.0**
Indiana University, Bloomington, IN, USA **2017-2019**

Bachelor of Technology (B.Tech) in Computer Science & Engineering, **I Division**
Dr. A.P.J. Abdul Kalam Technical University, INDIA **2007-2011**

Publications

Peer-Reviewed Journal Articles

10. **Parichit Sharma**, Marcin M. Stanislaw, Hasan Kurban, Oguzhan Kulekci, and Mehmet Dalkilic. "Geometric-k-means: A Bound Free Approach to Fast and Eco-Friendly k-means,", Machine Learning Journal, Springer, 2026.

9. Samir Abdaljalil, Hasan Kurban, **Parichit Sharma**, Erchin Serpedin, Rachad Atat "SINdex: Semantic INconsistency Index for Hallucination Detection in LLMs," *IEEE Open Journal of the Computer Society* 2026. (Accepted)

8. Hasan Kurban, **Parichit Sharma**, Mustafa Kurban, and Mehmet Dalkilic. "Accelerating Density of States Prediction in Zn-Doped MgO Nanoparticles via Kernel-Optimized Weighted k-NN," *Nature Scientific Reports, Springer*, 2025

7. **Parichit Sharma**, Sarthak Mishra, Hasan Kurban, Mehmet M. Dalkilic. "p-ClustVal: A Novel p-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data, "*International Journal of Data Science and Analytics, Springer*, 2025.

6. Mert O. Cakiroglu, Hasan Kurban, **Parichit Sharma**, M. Oguzhan Kulekci, Elham K. Buxton, Maryam R. Sarmazdeh, and Mehmet Dalkilic. "An extended de Bruijn graph for feature engineering over biological sequential data," *Machine Learning: Science and Technology*, 2024.

5. **Parichit Sharma**, Hasan Kurban, and Mehmet M. Dalkilic. "DCEM: An R package for Clustering Big Data via Data-centric Modification of Expectation Maximization," *SoftwareX*, 17, 100944, 2022.

4. Hasan Kurban, Mustafa Kurban, **Parichit Sharma**, and Mehmet M. Dalkilic. "Predicting Atom Types of Anatase TiO2 Nanoparticles with Machine Learning," *Key Engineering Materials*, vol.880, pp.89-94, 2021.

3. Ankit A Roy, Abhilesh S Dhawanjewar, **Parichit Sharma**, Gulzar Singh, and MS Madhusudhan. "Protein Interaction Z Score Assessment (PIZSA): an empirical scoring scheme for evaluation of protein–protein interactions," *Nucleic Acids Research*, vol.47, pp.331-337, 2019.

2. **Parichit Sharma**, Shrikant S. Mantri. "WImpiBLAST: Web interface for mpiBLAST to help biologists perform large-scale annotation using high performance computing," *PLOS ONE*, 2014.

Peer-Reviewed Conference Proceedings

10. Abdaljalil, S., **Sharma, P.**, Serpedin, E. and Kurban, H., 2026. Halluverse-M^3: A multitask multilingual benchmark for hallucination in LLMs. arXiv preprint arXiv:2602.06920. (In review, IJCAI 2026)

9. Instance-Based Learning-Driven Density of States Analysis in Functionalized Fullerene Derivatives for Optimizing Organic Photovoltaics, **Parichit Sharma**, Hasan Kurban, Mehmet Dalkilic, Mustafa Kurban, *IEEE International Conference on Compatibility, Power Electronics and Power Engineering, CPE, Antalya, Turkey*, 2025.

8. **Parichit Sharma**, Marcin M. Stanislaw, Hasan Kurban, Oguzhan Kulekci, and Mehmet Dalkilic. "Geometric-k-means: A Bound Free Approach to Fast and Eco-Friendly k-means," *arXiv:2508.06353*, 2025

7. p-ClustVal: A Novel p-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data (Extended Abstract), **Parichit Sharma**, Sarthak Mishra, Hasan Kurban, Mehmet M. Dalkilic, *The 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2024)*, San Diego, USA, 2024.

6. What Data-Centric AI Can Do For k-means: a Faster, Robust k-means-d, **Parichit Sharma**, Hasan Kurban and Mehmet M. Dalkilic, *The 41st International Conference on Machine Learning (ICML), Data-centric Machine Learning Research (DMLR): Datasets for Foundation Models*, Vienna, Austria, 2024.

5. Novel NBA Fantasy League driven by Engineered Team Chemistry and Scaled Position Statistics, Ganesh Arkanath, Nishad Gupta, Hasan Kurban, **Parichit Sharma**, Madhavan K R, Elham Khorasani Buxton and Mehmet M. Dalkilic, *IEEE International Conference on Big Data: Data-Centric AI*, Sorrento, Italy, 2023

4. Are Sports Awards About Sports? Using AI to Find the Answer, Anshumaan Shankar, Gowtham Veerabadran Rajasekaran, Jacob Hendricks, Jared Andrew Schlak, **Parichit Sharma**, Madhavan K R, Hasan Kurban and Mehmet M. Dalkilic, *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML/PKDD): 10th Workshop on Machine Learning and Data Mining for Sports Analytics*, Turin, Italy, 2023

3. AReS: An AutoML Regression Service for Data Analytics and Novel Data-centric Visualizations, Josh Elms, Sam Johnson, Madhavan K R, Keerthana Sugasi, **Parichit Sharma**, Hasan Kurban and Mehmet M. Dalkilic, *29th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Undergraduate Consortium (KDD-UC)*, Long Beach, CA, USA, 2023

2. Are They What They Claim: A Comprehensive Study of Ordinary Linear Regression Among the Top Machine Learning Libraries in Python, Sam Johnson, Josh Elms, Madhavan K R, Keerthana Sugasi, **Parichit Sharma**, Hasan Kurban and Mehmet M. Dalkilic, *29th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Undergraduate Consortium, (KDD-UC)*, Long Beach, CA, USA, 2023

1. Data Expressiveness and Its Use in Data-centric AI, Hasan Kurban, **Parichit Sharma** and Mehmet M. Dalkilic, *35th Conference on Neural Information Processing Systems (NeurIPS 2021) Data-centric AI, Sydney, Australia*, 2021

Published Open Source Software (Selected)

- **DCEM - An R package for soft-clustering of big data via a data-centric expectation maximization algorithm (downloads: 35K+)**: Developed a novel R package implementing EM* (EM-star), a data-centric modification of the traditional Expectation Maximization algorithm for clustering big data. This data-centric approach enables efficient processing of datasets with high dimensions, large sizes, and complex separability that overwhelm traditional iterative algorithms. Available on CRAN with over 30K downloads. **[Code]**
- **Data-Centric k-means - An open source API for clustering big data efficiently**: This package addresses critical scalability challenges in unsupervised machine learning by providing efficient clustering solutions for large-scale datasets. Available on PyPI for widespread community adoption, the tool facilitates high-performance clustering analysis in fields ranging from genomics and bioinformatics to market segmentation and pattern recognition. **[Code]**
- **pClustVal: a novel data transformation to enhance clustering of high dimensional single cell rna seq (scRNASeq) data**: Introduce a p-adic data transformation to address critical challenges in genomics data analysis by reducing cluster overlap and enhancing separation without requiring ground truth labels, making it broadly applicable across different clustering algorithms and dimensionality reduction techniques. **[Code]**
- **CLICK - A tool for topology independent comparison of biomolecular (proteins, RNA) 3D structures (Average Number of users: 100/month)**: CLICK employs a novel clique-based algorithm that identifies locally similar structural regions to establish global alignments, enabling detection of structural similarity even when chain topology differs significantly. Available as a freely accessible web platform with integrated visualization capabilities through JMol viewer. **[Code]**
- **PIZSA - A versatile web portal for studying protein interactions**: The PIZSA web server identify stable protein-protein interactions from among thousands of possible configurations, offering significant improvements over competing methods. Available as both web server and standalone software for widespread community adoption. **[Code]**
- **PACKPRED - A web server for studying the effect of point mutations in proteins**: Packpred employs the residue environment and packing in making the predictions. **[Code]**
- **WIMPIBLAST - A scalable platform for large scale genome annotation using high performance computing**: With next-generation sequencing technologies generating massive amounts of data, traditional sequence similarity searches can take days to complete on standard computers. WImpiBLAST harnesses the speed of mpiBLAST (a parallelized version of the popular BLAST sequence analysis tool) without needing expertise in Linux command line or HPC cluster operations. The use cases demonstrate remarkable improvements - in some instances achieving up to 95% faster results compared to standard searches. **[Code]**

Leadership Experience

- **Teaching Leadership: Co-lead** a team of 25+ graduate assistants (including PhD students) for an undergraduate programming course. **Created the teaching material and lab assignments** for the full 16-week semester. **Taught** the labs and conducted weekly office hours for the full 16 week semester. Ensured synchronization within the course team by **leading** weekly meetings throughout the semester. **Supervised** the development of the autograding software for ensuring consistency and scaling the grading to **450+ students**. My contributions played the key role to help increase the student enrollment by over 60% by the end of the 3^{rd} year of my leadership in the course.

- **Research Leadership:** Created the agenda for **5** research projects as an **independent researcher**. Collaborated with both **technical and non-technical stakeholders** for resource allocation. Executed the project plan, evaluated and benchmarked the results, leading to the publication of 7 original research articles.

- **Mentorship Experience:** Mentored **2 Ph.D., 7 M.S. and 2 B.S.** students to define their research project and streamline objectives. Provided **guidance** on designing the experiments and evaluating the results. **co-authored** the publication of 5 original research articles with the mentored students.

Accolades

- **Letter of Appreciation 2015, issued by the Head of the Department, Bioinformatics Lab, NABI, India**
  I provided the technical and computational support to the scientists in the lab. The letter was issued by the head of the Bioinformatics Lab, NABI, India - acknowledging my services and support for the research and contribution to the open-source projects.

- **Manthan Award 2013 for Onama in the E-Science & Technology Category, Government of India National Award for Digital Innovation (PRESS)**
  I served as one of the lead developer on this product. ONAMA addresses the critical national shortage of High Performance Computing (HPC) professionals by providing an accessible, cost-effective platform for engineering education and research.

- **International Data Corporation (IDC), HPC Innovation Excellence Award, 2011 (PRESS)**
  I was one of the Lead Developers for CHReME (C-DAC HPC Resource Management Engine), an indigenously developed web-based HPC resource management system that received the prestigious International Data Corporation (IDC) HPC Innovation Excellence Award at Supercomputing Conference 2011, Seattle. This award represents India's technological excellence in indigenous HPC software development and its global impact on advancing scientific computing capabilities.

Academic Service

Conference Sessions & Reviewing

- **[By Invitation of Conference Chairs]** Program Committee Member for the Society for Industrial & Applied Mathematics (SIAM) International Conference on Data Mining (SDM25).

- **[By Invitation of Conference Chairs]** Session Chair of the 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA), San Diego, United States, 2024.

- Program Committee Member for the 41st, International Conference on Machine Learning (ICML, 2024).

- Program Committee Member of the 41st, International Conference on Association for Advancement of Artificial Intelligence (AAAI, 2024).

- Program Committee Member of the 26th & 27th, International Conference on Artificial Intelligence and Statistics (AISTATS, 2023 & 2024).

Invited Reviewer for Journals I have also been a peer reviewer for **Nature Scientific Reports**, **International Journal of Data Science & Analytics, Springer** and PeerJ Computer Science.

Invited Talks

Conference & Workshops

- Instance-Based Learning-Driven Density of States Analysis in Functionalized Fullerene Derivatives for Optimizing Organic Photovoltaics, *IEEE International Conference on Compatibility, Power Electronics and Power Engineering, CPE, Antalya, Turkey, 2025*. **Main track presentation**
- p-ClustVal: A Novel p-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data, *11th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2024), San Diego, USA, 2024*. **Main track presentation**
- What Data-Centric AI Can Do For k-means: a Faster, Robust k-means-d, *41st International Conference on Machine Learning (ICML), Data-centric Machine Learning Research (DMLR): Datasets for Foundation Models, Vienna, Austria, 2024*. **Poster**
- Novel NBA Fantasy League driven by Engineered Team Chemistry and Scaled Position Statistics, *IEEE International Conference on Big Data: Data-Centric AI, Sorrento, Italy, December 15, 2023*. **Poster**
- Are Sports Awards About Sports? Using AI to Find the Answer, *ECML/PKDD: 10th Workshop on Machine Learning and Data Mining for Sports Analytics, Turin, Italy September 18, 2023*. **Workshop presentation**
- Data Expressiveness & Its Use in Data-Centric AI, *1st Data-Centric AI Workshop, 35th Conference on Neural Information Processing Systems (NeurIPS 2021), Sydney, Australia*. **Workshop presentation**

Teaching Experience

I have had the privilege of teaching and assisting with various courses, each presenting unique opportunities for growth as an educator. Among these, my role in the introductory programming course (CSCI-C200) demonstrates my commitment to excellence in computer science education and my ability to scale educational impact while maintaining quality. I helped transform C200 into a dynamic, cross-disciplinary learning environment **serving over 2500 students**, over 5 years, from diverse academic backgrounds including Physics, Mathematics, and Sociology, to name a few.

Instructor, Computer Science Department — **#Students: 8 (small class size)**

I taught the lectures for the full 8-week course (note: summer courses are intensive due to the short time period). **Supervised** the graduate assistants to organize and conduct weekly labs. **Helped** students to achieve **classroom success** by monitoring progress via quizzes and exams. Conducted one-to-one office hours for continued support.

- CSCI-C 200: Laboratory: Computation (Undergraduate) — Summer 2023
- CSCI-C 200: Introduction to Computers & Programming in Python (Undergraduate) — Summer 2023

Lead Associate Instructor, Computer Science Department — **#Students: 250-500**

I played the key role in **course organization and success**, and **managed a team of 15-35 graduate assistants** composed of BS, MS and PhD students. Established **well defined team and responsibility structure** to ensure timely delivery of course and laboratory material. **Supervised** the development of our own autograding software to ensure consistent grading for **500+** students. Provided online help for students by participating in Canvas's InScribe Q-n-A platform.

- CSCI-C 200: Introduction to Computers & Programming in Python (Undergraduate level) — Spring 2025, Fall 2024, Spring, 2024, Fall 2021, Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025.

Associate Instructor, Computer Science Department **Indiana University, USA** **#Students: 150-250**

I taught labs, conducted office hours, proctored and graded exams and course assignments. I also interacted with the students both offline (in office hours) and online (via Piazza) to answer questions and ensure student success.

- CSCI-P 536: Advanced Operating Systems (Graduate Level) — Spring 2019
- CSCI-B 365: Data Mining (Undergraduate Level) — Spring 2018
- CSCI-C 200: Introduction to Computers & Programming in Python (Undergraduate) — Fall 2019, Spring 2020, Fall 2020, Spring 2021

Professional Experience

University of Maryland, Baltimore, MD, USA
AI & ML, Postdoctoral Research Fellow, School of Medicine November 2025 - Present

- Building an end-to-end AI pipeline for label-free biomarker classification from nanopore electrophysiology data, leveraging ML and AI models trained on 50+ biophysical features including transition matrices, dwell times, and state occupancy probabilities.
- Establishing an AI/ML in Biology Special Interest Group (SIG) at UMB–UMCP, bridging the computational and biophysics communities through seminars, workshops, and cross-institutional collaboration in support of K99/R00 pathway funding.

Indiana University Bloomington, IN, USA
AI & Machine Learning, Research Assistant, Computer Science Department 2019 - 2025

- Conceptualized and implemented a novel data-centric k-means algorithm for big data, reducing the clustering time and computations by upto **99%**, while conserving the clustering partitions; published in ICML.
- Engineered an optimized Expectation-Maximization (EM) algorithm for soft clustering, improved resource usage efficiency by **50%**, while maintaining similar accuracy as the classical EM; published in NEURIPS.
- Open-sourced DCEM: an R package achieving **50X** faster convergence when clustering large-scale data **(2M+ points)**; validated framework across varying data dimensionality and cluster sizes (20-100 clusters) with comprehensive benchmarks; published in SoftwareX, Elsevier.

Computational Biology, Research Assistant, Computer Science Department

- Developed a scalable data transformation method for enhanced clustering of **Single Cell RNA Sequencing data**, improving tissue identification in **91%** of trials, across **30 experiments with 1200+ observations**; published in International Journal of Data Science & Analytics, Springer.
- Created unsupervised parameter estimation achieving up to **84%** accuracy improvement in specific tissue identification, on **highly sparse (87%) scRNASeq data**, consistently outperforming baseline methods across 5 machine learning algorithms.
- Benchmarked **60+ machine learning models** for identifying motifs that can act as successful predictors of TIMP sequences, improved motif classification by **20%** over state-of-the-art protein language; published in Machine Learning: Science & Technology.

Texas A&M University at Qatar, Doha, Qatar
Research Associate, Electrical & Computer Engineering Department May 2024 - July, 2024

- Contributed to developing a novel pipeline for automated hallucination detection in large language models (LLMs) achieving 60X speedup and improved accuracy by 9.4% over SOTA LLMs.

Ancestry Inc, Utah, USA

Bioinformatics Intern, DNA Science Team May 2019 - August, 2019

- Implemented a data analysis protocol for cleaning, preprocessing and analyzing **80 million** genealogical records of Ancestry customers, improving genetic community annotation via statistical hypothesis testing and recommending new document collections for Ancestry's customers.

Google Summer of Code, Python Software Foundation, USA

MRI Imaging & Analysis, Diffusion Imaging in Python (DIPY) May 2018 - August 2018

- Developed user-friendly workflows for neuro-image (MRI-Scan) data pre-processing, analysis, and registration (affine and diffeomorphic).

Indian Institute of Science Education and Research (IISER), Pune, India

Scientific Programmer, Computational & Structural Biology Lab 2015 - 2017

- Optimized 3D least squares algorithm for aligning protein structures, reducing protein database search time from **10 to 2 hours**.
- Collaborated with domain scientists and architected a scalable web platform for **processing, aligning and visualizing biomolecular structures**, integrated key statistical insights, while visualizing results in a user-friendly web portal.
- Designed PIZSA – an open-source web platform for protein interactions, outperforming baseline algorithms in **69%** of cases; published in Nucleic Acids Research (NAR).
- Developed an end-to-end pipeline for genotyping the HLA data from **next generation sequencing** platform, achieving **95%** accuracy through consensus clustering of sequences.

National Agri-Food Biotechnology Institute, Mohali, India

Application Engineer (on deputation from C-DAC), Bioinformatics & Genomics Lab 2013 - 2015

- Created a high-performance web portal for scientists to search 10M+ genomics sequences, reducing search time from **96 to 4 hours** via distributed searches on high performance computing cluster; published in PlosOne.
- Collaborated with domain scientists to identify and resolve bottleneck in local genome sequence searches, reduced manual intervention time from **5 hours to 10 minutes** by deploying light-weight genome search servers on in-house linux severs.

Center for Development of Advanced Computing, Pune, India

Consultant, High Performance Computing Solutions Group 2012 - 2014

- Contributed to the development of CHReME, developed modules to manage computing resources on a distributed high performance computing (HPC) cluster.
- Contributed in the development of Onama, designed Linux based automation tools for scientific application installation, and management by HPC system admins.

Technical Skills

- **ML:** Supervised Learning, Unsupervised Learning, Self-Supervised Learning, Dimension Reduction
- **AI Fraemworks:** TensorFlow, Keras, Scikit-learn, Pandas, scikit
- **Programming Languages:** Python, R, Linux Bash Scripting
- **Software Development:** Visual Studio Code (VSC), GIT
- **Bioinformatics:** End-to-End Pipeline development, Method development for analyzing Single Cell Data, Algorithm for processing

transcriptomics data

- **Biological Datasets:** Next Generation Sequencing data (NGS), Genomics, Single cell RNA sequencing, 3D Protein Structure data.
- **Web Development:** Django, HTML5, CSS3, AJAX, Javascript, Jquery
- **Infrastructure:** High Performance Cluster (HPC), Linux, Windows

(References available upon request)